Spectral Active Learning with Dynamic Model Management for Uncertainty Quantification of Computationally Expensive Black-Box Simulators

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https://github.com/Yugam2508/fyp-active-learning-cfd

Abstract

Uncertainty quantification (UQ) for high-fidelity computational fluid dynamics (CFD) simulations of hazardous gas dispersion is constrained by a severe budget mismatch: each run costs hours of wall-clock time, while classical Monte Carlo UQ needs thousands of evaluations. Surrogate models bridge this gap, but we identify and characterise a specific catastrophic failure mode of the standard tool—Gaussian Process (GP) regression—under which it is unsafe to deploy passively. We show experimentally that GP *prior reversion* produces near-constant, physically meaningless predictions only when *two* conditions hold simultaneously: a large input domain relative to the sampling budget, *and* a high output dynamic range. A domain-size ablation locates a sharp crossover between half-widths 2 and 10, and a Schwefel control (ρ=1.09) confirms that large domain size alone is insufficient. To recover performance within the same budget we propose an active learning framework combining a heterogeneous surrogate committee, a spectral uncertainty acquisition metric (UEigV) derived from the normalised graph Laplacian of the committee prediction graph, and a consistency-based model manager available in a hard-rejection (LCS) and a soft, reversible cross-validation-weighted variant. On the target regime the framework reduces RMSE 23-fold at an equal budget of 20 evaluations. We then complete the pipeline with a forward-UQ stage that propagates the input distribution through the trained committee to deliver an output probability box, bounded exceedance probabilities, a law-of-total-variance epistemic/aleatoric decomposition that doubles as a convergence diagnostic, and Sobol sensitivity indices with epistemic error bars—all validated against ground-truth Monte Carlo. We give an honest delineation of where the framework wins and where a passive GP suffices.

1 Introduction

The transition toward low-carbon energy has renewed interest in ammonia as a hydrogen carrier, but ammonia is toxic and accidental releases form dense, ground-hugging clouds whose spatial concentration must be predicted for quantitative risk assessment [1]. CFD is the industry standard for such dispersion modelling, solving the Navier–Stokes equations coupled with species transport, but a single high-resolution run can take hours to days on HPC infrastructure. This expense is a fundamental barrier to UQ: Monte Carlo (MC) converges as $N^{-1/2}$ [2], so reducing error tenfold needs 100× more runs, while the feasible CFD budget is often fewer than twenty evaluations.

Surrogate models—cheap emulators trained on a handful of high-fidelity evaluations—are the standard remedy [3, 4], with GP regression the workhorse for computer experiments [5, 6]. A surrogate that is overconfident in data-sparse regions, however, contributes model-form uncertainty invisible to its own variance estimate [7]. We make this concrete by characterising *GP prior reversion*: when the average inter-sample distance exceeds the learnable kernel length scale, GP predictions collapse to the prior mean. For high-dynamic-range responses—like CFD concentration fields spanning orders of magnitude—this prior mean is catastrophically wrong.

This paper makes four contributions. **(1)** A falsifiable, two-condition criterion (large domain × high dynamic range) for when passive GP surrogates fail catastrophically, supported by a domain-size ablation and a Schwefel control. **(2)** A spectral active learning framework whose acquisition metric, UEigV, transfers the graph-Laplacian uncertainty measure of [8] from language-model UQ to physical surrogate modelling for the first time. **(3)** A consistency-based model manager in two forms—hard Largest Consistent Subset (LCS) rejection and a soft, reversible cross-validation-weighted variant—and an analysis of why naive soft schemes fail. **(4)** A forward-UQ stage that turns the trained committee into a Monte Carlo engine delivering a probability box, bounded exceedance probabilities, a variance decomposition that serves as a stopping diagnostic, and Sobol indices with epistemic bars.

2 Failure Characterisation

For a GP with prior mean μ and kernel k, the posterior predictive mean at a test point $\mathbf{x}^*$ is

$$\hat{f}(\mathbf{x}^*) = \mu + \mathbf{k}_*^\top \left(\mathbf{K} + \sigma_\varepsilon^2 \mathbf{I}\right)^{-1} (\mathbf{y} - \mu \mathbf{1}). \tag{1}$$

As the minimum distance between $\mathbf{x}^*$ and the training set grows relative to the length scale ℓ, the RBF covariance $k(\mathbf{x}^*, \mathbf{x}_i) = \sigma_f^2 \exp(-\|\mathbf{x}^* - \mathbf{x}_i\|^2 / 2\ell^2) \to 0$, so

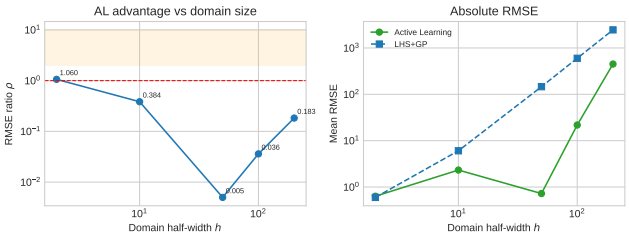


Figure 1: Domain-size ablation on the physics benchmark (30 trials, N=20 per method). Left: RMSE ratio ρ versus half-width; the shaded band is the crossover zone. Right: absolute mean RMSE; the trajectories diverge as the domain grows.

$\hat{f} \to \mu$ regardless of the data. On the physics benchmark $f_{\text{phys}}(\mathbf{x}) = \sin(2x_1)\cos(2x_2) + e^{-x_3^2} + 0.2\|\mathbf{x}\|^2$ over $[-200, 200]^3$, with 20 Latin-Hypercube (LHS) samples the expected nearest-neighbour distance is $\mathcal{O}(100)$ per dimension; the marginal likelihood cannot identify a length scale long enough to bridge the gaps, and the GP reverts to its standardised prior mean across most of the domain.

Domain-size ablation. Figure 1 evaluates the benchmark at half-widths $h \in \{2, 10, 50, 100, 200\}$ with all other settings fixed. A sharp crossover appears between h=2 (ρ=1.06, indistinguishable) and h=10 (ρ=0.384), where $\rho = \text{RMSE}_{\text{AL}}/\text{RMSE}_{\text{LHS+GP}}$. The peak advantage ρ=0.005 (a 200× reduction) occurs at h=50.

The Schwefel control. The Schwefel function on $[-500, 500]^3$ has a *larger* domain than the physics benchmark, yet active learning does not win (ρ=1.09). Its output varies only between $\pm 418d$ with the minimum near zero, so the GP prior mean is a reasonable approximation even in data-sparse regions. This isolates the mechanism: catastrophic failure requires large domain size *and* high output dynamic range. This two-condition criterion is the central empirical finding.

3 Spectral Active Learning Framework

Heterogeneous committee. A committee of up to K=9 surrogates spans complementary inductive biases: two GPs (RBF and Matérn-3/2), two Polynomial Chaos Expansions (PCE, degrees 2 and 3, Bayesian Ridge), two Random Forests (50/100 trees), an RBF Support Vector Regressor, and two dropout DNNs (2×64, 2×128). Structural heterogeneity is essential because identical models share a failure mode and reveal no model-form uncertainty [9].

UEigV acquisition. At a candidate $\mathbf{x}$ the K predictions form nodes of a fully connected graph with RBF

Algorithm 1 Spectral Active Learning with model management

Require: domain $\mathcal{X}$, budget $N_{\max}$, N_0=5, threshold α
1: $\mathcal{D}_0 \leftarrow \text{LHS}(N_0, \mathcal{X})$; train committee on $\mathcal{D}_0$
2: **for** $t = 1, \ldots, N_{\max} - N_0$ **do**
3: sample candidate pool $\mathcal{X}_c$ (proximity-filtered)
4: $\mathbf{x}^* \leftarrow \arg\max_{\mathbf{x} \in \mathcal{X}_c} \text{UEigV}(\mathbf{x})$
5: $y^* \leftarrow f(\mathbf{x}^*)$; $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\mathbf{x}^*, y^*)\}$
6: retrain all active models on $\mathcal{D}_t$
7: apply model manager (hard LCS *or* soft CV weights)
8: **end for**
9: **return** committee, $\mathcal{D}_{N_{\max}}$

edge weights $A_{ij} = \exp(-(\hat{y}_i - \hat{y}_j)^2/2\sigma^2)$. With normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ and eigenvalues $\lambda_k \in [0, 2]$, the acquisition score is

$$\text{UEigV}(\mathbf{x}) = \sum_{k=1}^{K} \max(0,\, 1 - \lambda_k)\,, \tag{2}$$

a continuous relaxation of the number of agreement clusters [8]. Full consensus gives UEigV $\approx$ 0; a two-way split gives $\approx$ 2. The next query maximises UEigV over a proximity-filtered random candidate pool. Different families extrapolate differently in data-sparse regions—a GP reverts to its mean, a PCE follows the global polynomial, an RF returns leaf means—so UEigV is high precisely where the function is least constrained.

Model management. Not all surrogates learn at the same rate; high-capacity DNNs overfit early and inject phantom disagreement into UEigV. The *hard* LCS manager scores each model by its reduced chi-square deviation from the ensemble mean and permanently rejects the worst if it exceeds the α-quantile threshold (with a minimum-data guard and a floor of two active models). We use the ensemble mean, not the true values, as the reference: every surrogate interpolates the sparse training data, so training-point residuals against truth are uninformative; relative consistency is the criterion that discriminates in the active-learning regime. We set α=0.99, identified by sensitivity analysis as the best diversity–stability balance.

Soft, reversible weighting. Permanent rejection is irreversible: a DNN that is useless at N=8 but informative at N=16 is gone for good. We therefore offer a soft variant that weights rather than removes. The working scheme uses k-fold cross-validation error,

$$w_k \;\propto\; \frac{1}{\text{CV-MSE}_k\,(1 + \pi_k)}, \tag{3}$$

i.e. inverse-variance (weighted-average-surrogate) weighting [10, 11], with an optional physics penalty

Table 1: Cross-benchmark performance, N=20, 30 trials. $\rho = \overline{\text{RMSE}}_{\text{AL}}/\overline{\text{RMSE}}_{\text{LHS+GP}}$.

Problem	AL mean	LHS+GP	ρ
Physics $[-200, 200]^3$	111	2,504	**0.044**
Borehole (8-D)	15.06	13.57	1.110
Piston (7-D)	0.105	0.093	1.133
Friedman (5-D)	2.887	2.467	1.170
Nonstationary (3-D)	0.792	0.839	0.944
Branin (2-D)	23.57	20.46	1.152
Hartmann3 (3-D)	0.658	0.570	1.156
Ishigami (3-D)	3.941	3.605	1.093
Schwefel (3-D)	376	344	1.092

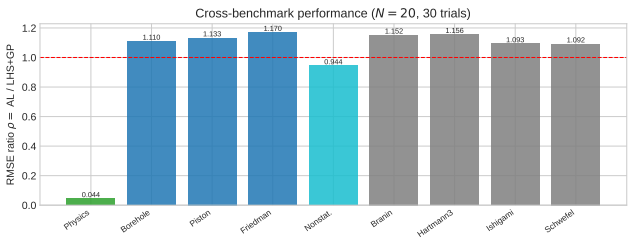


Figure 2: RMSE ratio ρ across nine benchmarks (N=20, 30 trials). Below the dashed break-even line, active learning wins. Only the physics benchmark (green) shows a decisive advantage.

$\pi_k \geq 0$ for violations of known output bounds (e.g. concentration non-negativity), in the spirit of physics-informed ML [12]. Models below a weight floor are *transiently* muted from the UEigV graph but keep retraining and may recover. Two natural alternatives *fail* and are instructive: (i) Boltzmann weights on the training-point chi-square score ($w_k \propto e^{-s_k/2T}$) leave catastrophic extrapolators near-uniformly weighted because the score cannot separate models that all interpolate the data, contaminating the mixture; (ii) consensus-deviation probe scores punish the single *correct* dissenter when the majority prior-reverts together. CV error sidesteps both by scoring against held-out data, separating members by orders of magnitude.

4 Cross-Benchmark Results

All methods use a fixed budget of N=20 (5 initial LHS + 15 steps) over 30 independent trials, with RMSE on a 5,000-point held-out set.

Table 1 and Fig. 2 delineate the operating regime honestly. The physics benchmark—the one problem satisfying both failure conditions—shows a decisive 23× RMSE reduction (ρ=0.044). The nonstationary benchmark is marginally favourable (ρ=0.944); on the remaining seven problems a well-specified GP ties or wins by 9–17%. The framework does *not* claim universal superiority; it claims a precise, testable criterion for when adaptive sampling is warranted.

Success mechanism and an honest caveat. Instrumenting the runs shows that the manager sequentially removes the misspecified majority and the surviving ensemble is anchored by PCE-d2—the exact model class for the dominant quadratic term, which accounts for $> 99.9\%$ of the output variance. The win is therefore best understood as consistency-based *model selection* (an anchor-model funnel) with UEigV supplying the disagreement signal that makes the selection identifiable, rather than a single interior "discovery" event. The two-condition criterion is thus necessary but not sufficient: the committee must also contain a surrogate whose hypothesis class overlaps the response's dominant structure. In a paired 30-trial comparison (identical seeds and initial designs), the soft CV-weighted variant reproduces this behaviour smoothly and reversibly—median RMSE 0.64 vs 0.71 for hard LCS, success rates 87% vs 90%, Wilcoxon signed-rank p=0.096: statistical parity in accuracy. Notably, only one failure seed is shared between the two methods, indicating that failures are driven by the stochastic query trajectory as well as by initial coverage.

5 Forward Uncertainty Quantification

The trained committee is a Monte Carlo engine: a 50,000-point study costs seconds on the surrogate versus ∼50,000 CFD-hours on the solver. We propagate the input distribution through all surrogates and report four quantities, each with epistemic bars from committee disagreement, validated against a 200,000-point ground-truth MC on the synthetic benchmark.

Output PDF and probability box. Figure 3 shows the CDF probability box: at each level y the committee CDFs span $[F_{\text{low}}(y), F_{\text{up}}(y)]$, a p-box in the sense of [13]. Exceedance probabilities follow as bounded intervals: for a 95th-percentile safety threshold, $P(Y > y_c)$ is returned as an interval that brackets the truth—the statement a risk assessment needs, rather than a single unjustified point estimate.

Variance decomposition as a stopping rule. By the law of total variance over the input distribution,

$$\text{Var}[Y] = \underbrace{\text{Var}_{\mathbf{x}}\big[\mu_w(\mathbf{x})\big]}_{\text{input-driven}} + \underbrace{\mathbb{E}_{\mathbf{x}}\big[\sigma^2_{\text{epi}}(\mathbf{x})\big]}_{\text{epistemic}} (+\sigma^2_\varepsilon). \quad (4)$$

The epistemic fraction (Fig. 4) is a convergence diagnostic: while it dominates, the next evaluation should buy another simulation; once it is small, the UQ estimate is committee-converged. In our run it decays from 0.76 to ≈ 0 over the budget. This answers the framework's open

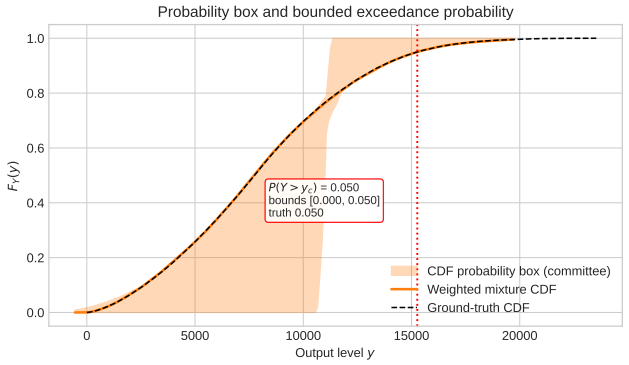


Figure 3: CDF probability box and bounded exceedance probability. The band is the committee envelope, the solid line the weighted-mixture CDF, the dashed line the ground-truth CDF. The annotated $P(Y > y_c)$ interval brackets truth.

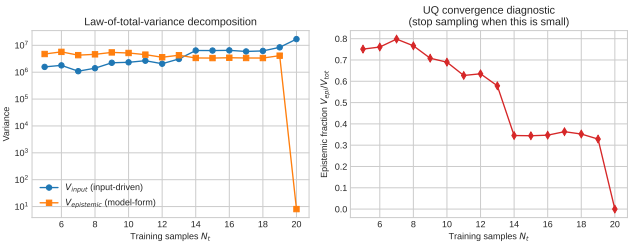


Figure 4: Law-of-total-variance decomposition versus sample count. Left: input-driven and epistemic variance. Right: the epistemic fraction, a convergence/stopping diagnostic.

problem of online stopping in directly interpretable UQ terms.

Sensitivity with epistemic bars. First-order and total Sobol indices are estimated by the Saltelli/Jansen pick-and-freeze scheme [14, 15] on the cheap ensemble, with per-model indices furnishing error bars on the sensitivity *ranking* (Fig. 5). The ensemble estimates track the ground-truth indices, closing the loop with classical dispersion sensitivity studies that obtained the same rankings only by expensive direct MC [16].

Central estimates are accurate for both managers: over 30 trials the median relative error of the predicted output median is 0.24% (successful runs), and a representative run matches truth to within 1% on mean, std, and all reported quantiles. The managers differ sharply, however, in *epistemic expressiveness* (Fig. 6): hard LCS collapses the committee to a single surviving model in every trial, so its epistemic bands are degenerate—the mean band and exceedance interval never cover the truth and the p-box has zero width. The soft variant retains all nine members with concentrated weights: its mean band covers the true output mean in 97% of trials and the p-box brackets 57% of the true CDF on average, though the exceedance interval under-covers in the far upper tail

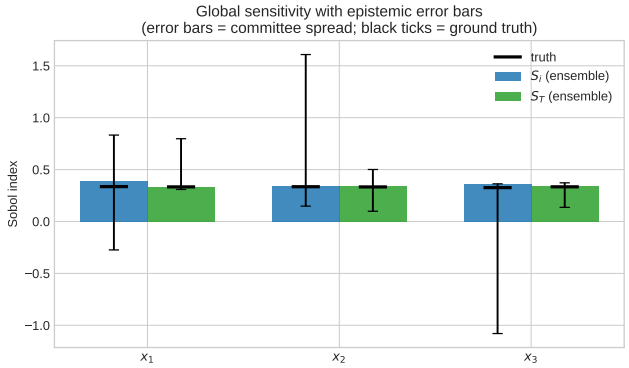


Figure 5: Sobol first-order and total indices of the weighted ensemble. Error bars are committee spread (epistemic); black ticks are ground-truth indices.

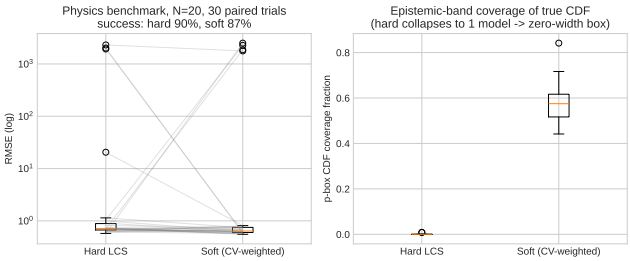


Figure 6: Paired 30-trial comparison. Left: RMSE distributions (grey lines join identical seeds); accuracy is at parity. Right: p-box coverage of the true CDF; hard rejection collapses to one model and loses all epistemic expressiveness, while soft CV weighting preserves it.

(13%), motivating conformal calibration as future work. Accuracy is at parity; the capacity to *quantify* model-form uncertainty after convergence belongs to the soft manager alone.

6 Discussion and Limitations

The benchmark is a smooth synthetic construction; real CFD responses include sharp gradients and recirculation that it does not replicate, so the observed improvements require validation on a live solver before safety-critical use. The epistemic bands are committee envelopes, not calibrated probabilities, and K=9 gives a coarse envelope—conformal prediction [17] is the principled upgrade. Cross-validation at $N \approx 20$ is itself noisy, so soft weights should be read as order-of-magnitude statements (which is all inverse-error weighting needs). Finally, the demonstrated advantage is partly attributable to PCE-d2 matching the benchmark's dominant term; the committee must contain a surrogate whose hypothesis class overlaps the true response. DNN surrogates are two-hidden-layer ReLU networks (Adam, 800 epochs, standardised I/O) implemented in PyTorch or, equivalently, scikit-learn's MLP; the released code supports

both backends.

7 Conclusion

Passive GP surrogates fail catastrophically precisely when the input domain is large enough to cause prior reversion *and* the response has high enough dynamic range to make the prior mean wrong. In that regime, 20 strategically selected evaluations achieve a 23× RMSE reduction over 20 passive ones via spectral disagreement among structurally diverse surrogates, stabilised by consistency-based model management—hard or soft. Completed with a forward-UQ stage delivering a probability box, bounded exceedance probabilities, a variance-decomposition stopping diagnostic, and Sobol indices with epistemic bars, the framework provides a principled and honestly delineated foundation for UQ of expensive black-box simulators, with CFD-based ammonia dispersion as the motivating application.

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